LAB ACTIVITY **10/08/2026**

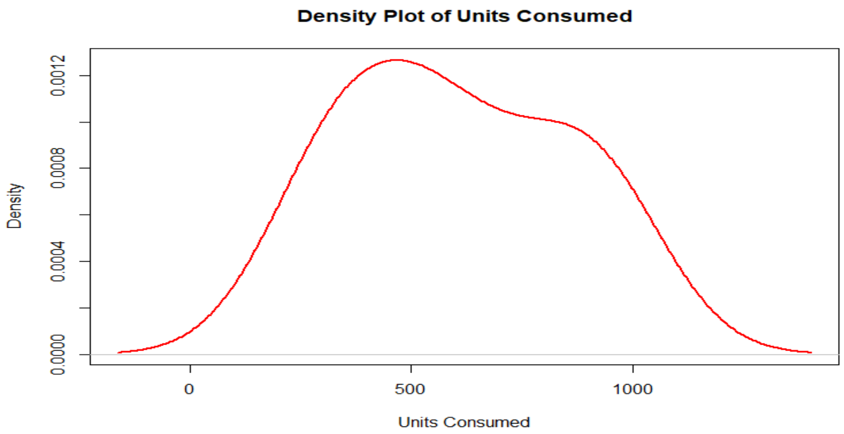
Set 21: Energy Consumption

1. Import the dataset in R. Plot a histogram and density plot for Units Consumed. Compare consumption patterns.

Program:

```
# Dataset
data <- data.frame(
 Sector=c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region=c("North","South","West","East","North","South"),
  Month=c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature=c(15,24,20,18,28,30),
  Units_Consumed=c(320,540,880,350,610,920),
  Cost=c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage=c(22,18,12,25,20,15),
  Peak_Hours=c(4,6,8,5,7,9)
)
# Histogram
hist(data$Units_Consumed, col="lightblue",
     main="Histogram of Units Consumed", xlab="Units Consumed")
# Density Plot
plot(density(data$Units_Consumed), col="red", lwd=2,
     main="Density Plot of Units Consumed", xlab="Units Consumed")
```

Output:

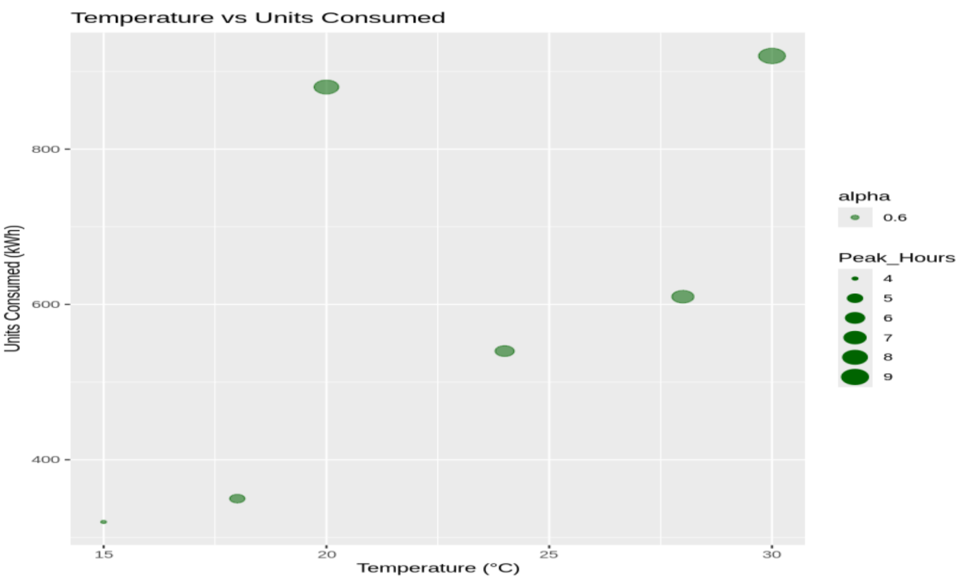


2. Using R programming, create a scatter plot between Temperature and Units Consumed. Represent Peak Hours using bubble size. Apply transparency.

Program:

```
data <- data.frame(
 Sector=c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region=c("North","South","West","East","North","South"),
  Month=c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature=c(15,24,20,18,28,30),
  Units_Consumed=c(320,540,880,350,610,920),
  Cost=c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage=c(22,18,12,25,20,15),
  Peak_Hours=c(4,6,8,5,7,9)
)
library(ggplot2)
ggplot(data, aes(x=Temperature, y=Units_Consumed,
                 size=Peak_Hours, alpha=0.6)) +
  geom_point(color="darkgreen") +
  labs(title="Temperature vs Units Consumed",
       x="Temperature (°C)", y="Units Consumed (kWh)")
```

Output:

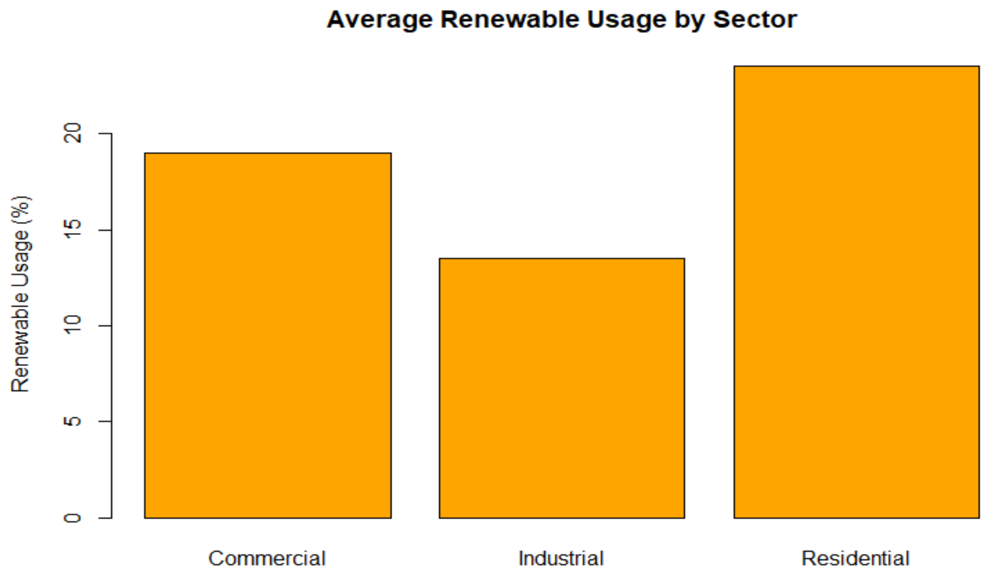


3. Using R, calculate average Renewable Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

Program:

```
data <- data.frame(
 Sector=c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region=c("North","South","West","East","North","South"),
  Month=c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature=c(15,24,20,18,28,30),
  Units_Consumed=c(320,540,880,350,610,920),
  Cost=c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage=c(22,18,12,25,20,15),
  Peak_Hours=c(4,6,8,5,7,9)
)
avg_renew <- aggregate(Renewable_Usage ~ Sector, data, mean)
barplot(avg_renew$Renewable_Usage, names.arg=avg_renew$Sector,
        col="orange", main="Average Renewable Usage by Sector",
        ylab="Renewable Usage (%)")
```

Output:



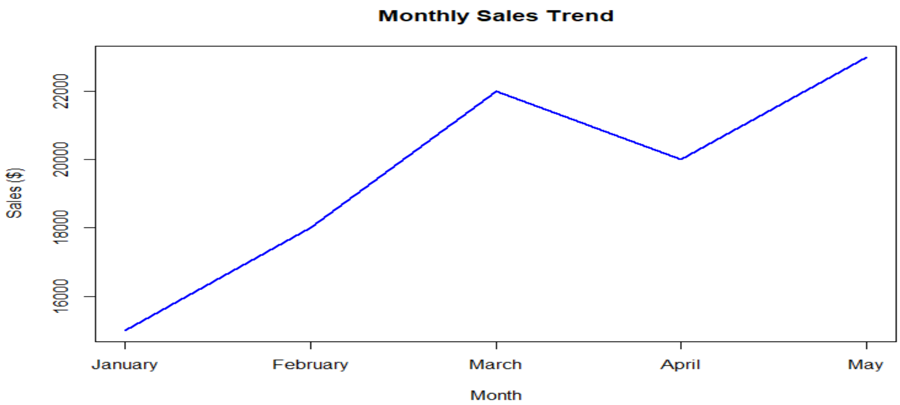
Set 22: Time Series Analysis

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

Program:

```
sales <- data.frame(
  Month=c("January","February","March","April","May"),
  Sales=c(15000,18000,22000,20000,23000)
)
plot(sales$Sales, type="l", col="blue", lwd=2,
     xaxt="n", xlab="Month", ylab="Sales ($)",
     main="Monthly Sales Trend")
axis(1, at=1:5, labels=sales$Month)
```

Output:

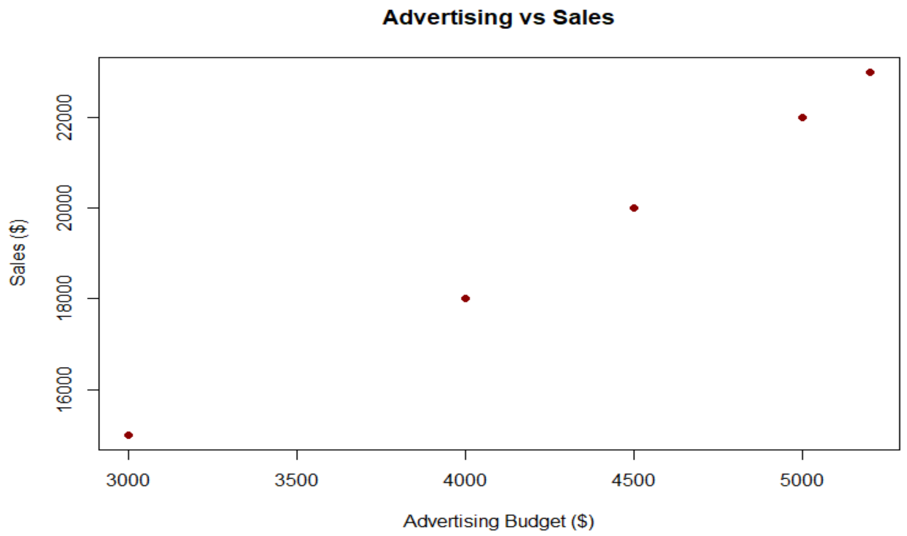


2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

Program:

```
sales <- data.frame(
  Month=c("January","February","March","April","May"),
  Sales=c(15000,18000,22000,20000,23000),
  Advertising_Budget=c(3000,4000,5000,4500,5200)
)
plot(sales$Advertising_Budget, sales$Sales,
     xlab="Advertising Budget ($)", ylab="Sales ($)",
     main="Advertising vs Sales", pch=19, col="darkred")
```

Output:

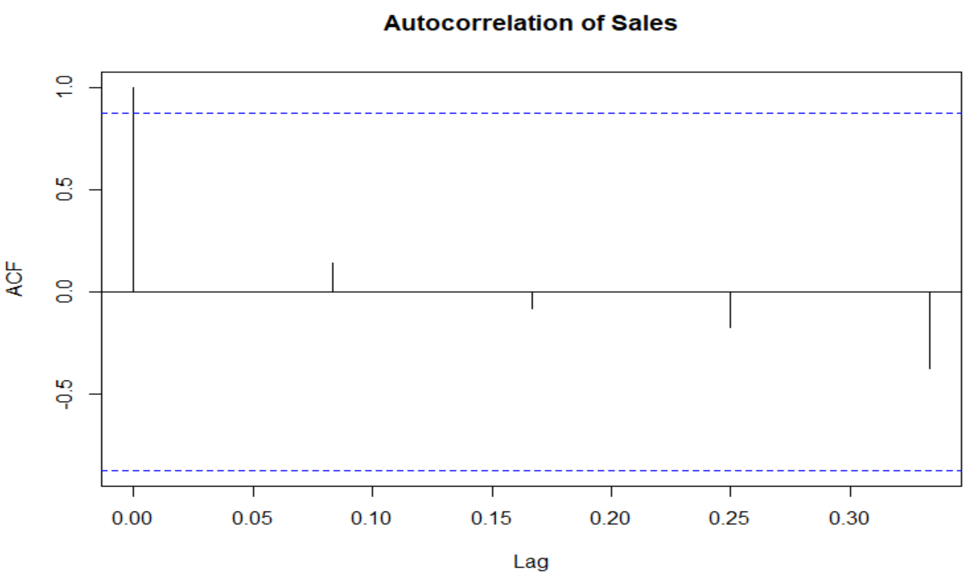


4. In R, create an autocorrelation plot to identify seasonality in the time series data.

Program:

```
sales <- data.frame(
  Month=c("January","February","March","April","May"),
  Sales=c(15000,18000,22000,20000,23000)
)
ts_sales <- ts(sales$Sales, frequency=12)
acf(ts_sales, main="Autocorrelation of Sales")
```

Output:



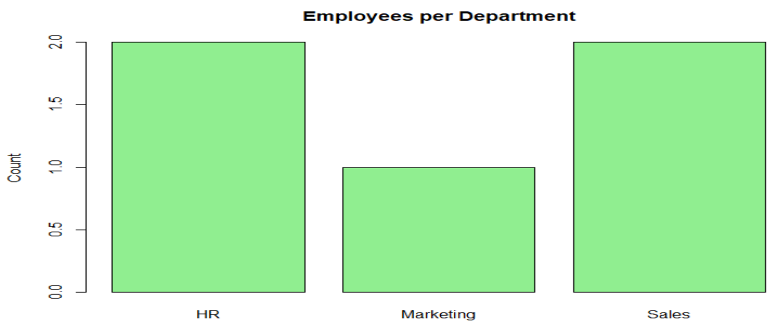
Set 23: Employee Performance Analysis

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

Program:

```
emp <- data.frame(
  EmployeeID=1:5,
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service=c(5,3,7,4,2),
  Performance_Score=c(85,92,78,90,76)
)
barplot(table(emp$Department), col="lightgreen",
        main="Employees per Department", ylab="Count")
```

Output:

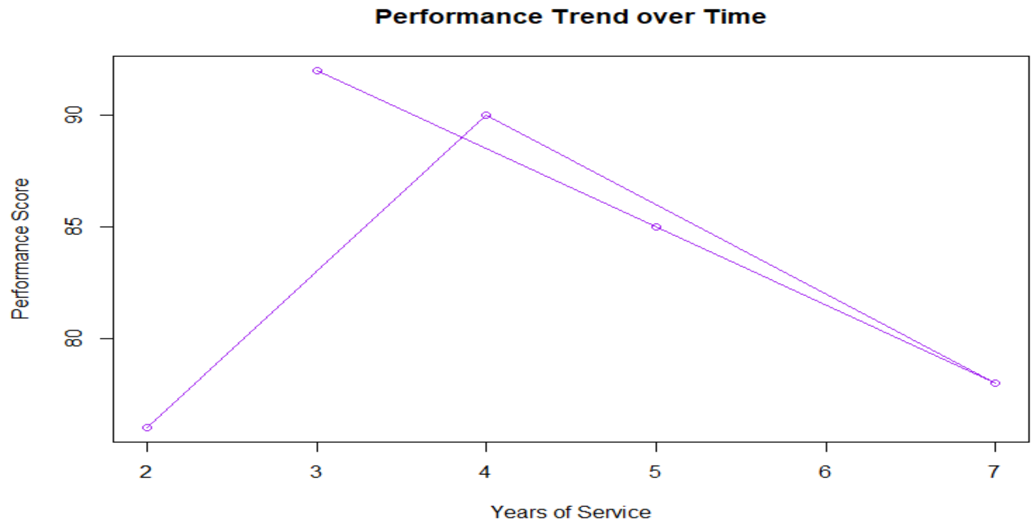


2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Program:

```
emp <- data.frame(
  EmployeeID=1:5,
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service=c(5,3,7,4,2),
  Performance_Score=c(85,92,78,90,76)
)
plot(emp$Years_of_Service, emp$Performance_Score, type="o", col="purple",
     xlab="Years of Service", ylab="Performance Score",
     main="Performance Trend over Time")
```

Output:



4. In R, develop a table showing the performance data for each employee. Label the table elements.

Program:

```
emp <- data.frame(
  EmployeeID=1:5,
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service=c(5,3,7,4,2),
  Performance_Score=c(85,92,78,90,76)
)
library(knitr)
kable(emp, caption="Employee Performance Data")
```

Output:

```
***

    Table: Employee Performance Data

    | EmployeeID|Department | Years_of_Service| Performance_Score|
    |----------:|:----------|----------------:|-----------------:|
    |          1|Sales      |                5|                85|
    |          2|HR         |                3|                92|
    |          3|Marketing  |                7|                78|
    |          4|Sales      |                4|                90|
    |          5|HR         |                2|                76|
```

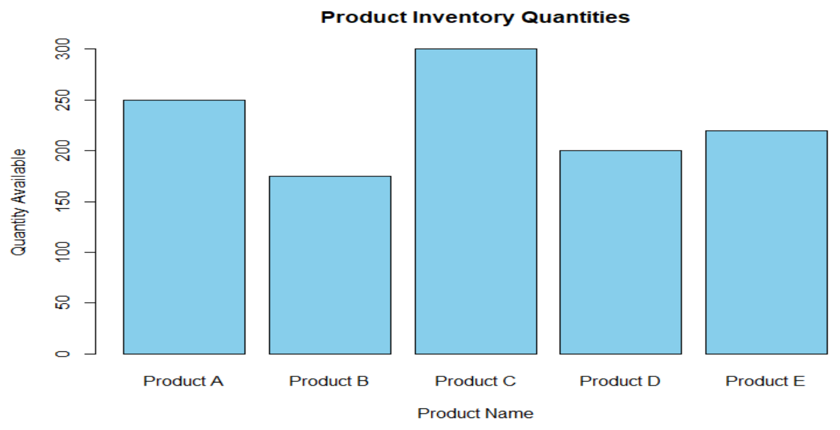
Set 24 Product Inventory Management

1.In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Program:

```
inventory <- data.frame(
  ProductID=1:5,
  ProductName=c("Product A","Product B","Product C","Product D","Product E"),
  Quantity=c(250,175,300,200,220)
)
barplot(inventory$Quantity, names.arg=inventory$ProductName,
        col="skyblue", main="Product Inventory Quantities",
        xlab="Product Name", ylab="Quantity Available")
```

Output:



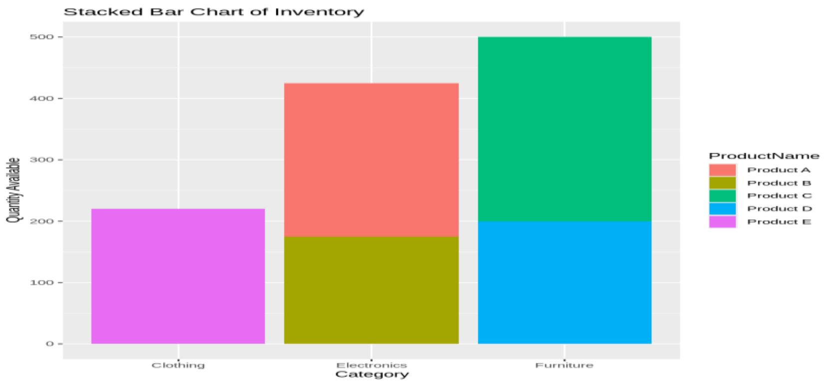
2.In R, generate a stacked bar chart to show the quantity of each product within different product categories.

Program:

```
inventory <- data.frame(
  ProductID=1:5,
  ProductName=c("Product A","Product B","Product C","Product D","Product E"),
  Quantity=c(250,175,300,200,220),
  Category=c("Electronics","Electronics","Furniture","Furniture","Clothing")
)
library(ggplot2)
```

```
ggplot(inventory, aes(x=Category, y=Quantity, fill=ProductName)) +
  geom_bar(stat="identity") +
  labs(title="Stacked Bar Chart of Inventory",
       x="Category", y="Quantity Available")
```

Output:

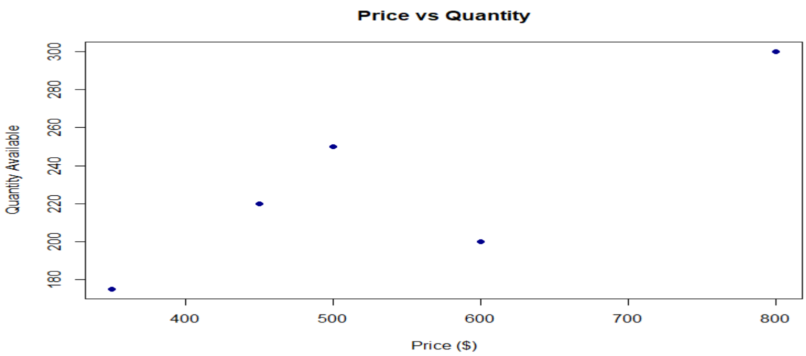


4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

Program:

```
inventory <- data.frame(
  ProductID=1:5,
  ProductName=c("Product A","Product B","Product C","Product D","Product E"),
  Quantity=c(250,175,300,200,220),
  Price=c(500,350,800,600,450)
)
plot(inventory$Price, inventory$Quantity,
     xlab="Price ($)", ylab="Quantity Available",
     main="Price vs Quantity", pch=19, col="darkblue")
```

Output:



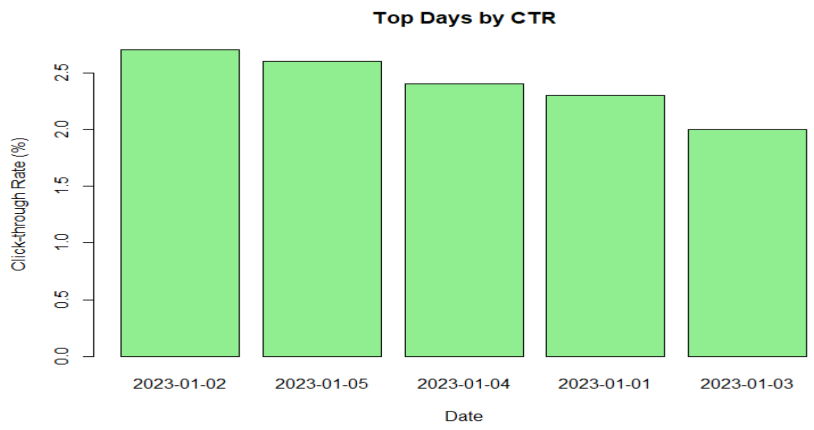
Set 25: Website Traffic Analysis

2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

Program:

```
traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  PageViews=c(1500,1600,1400,1650,1800),
  CTR=c(2.3,2.7,2.0,2.4,2.6)
)
# Order by CTR descending
traffic_sorted <- traffic[order(-traffic$CTR),]
barplot(traffic_sorted$CTR, names.arg=traffic_sorted$Date,
        col="lightgreen", main="Top Days by CTR",
        xlab="Date", ylab="Click-through Rate (%)")
```

Output:



3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

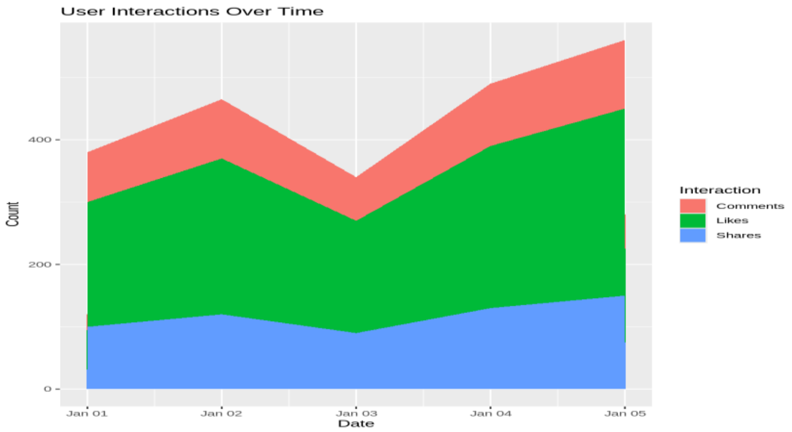
Program:

```
# Dataset
traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  Likes=c(200,250,180,260,300),
  Shares=c(100,120,90,130,150),
```

```
  Comments=c(80,95,70,100,110)
)
# Convert to long format
library(ggplot2)
library(tidyr)
traffic_long <- pivot_longer(traffic, cols=c(Likes, Shares, Comments),
                             names_to="Interaction", values_to="Count")

# Stacked area chart
ggplot(traffic_long, aes(x=Date, y=Count, fill=Interaction)) +
  geom_area() +
  labs(title="User Interactions Over Time",
       x="Date", y="Count")
```

Output:



SET 26 : Student Mini

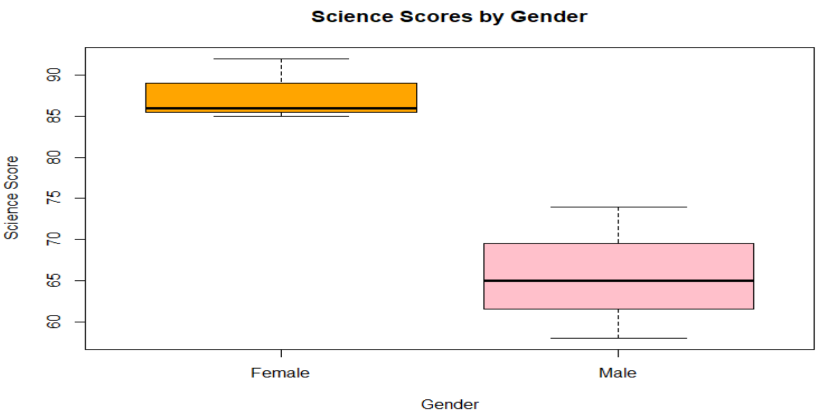
1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

Program:

```
students <- data.frame(
  StudentID=c("S01","S02","S03","S04","S05","S06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Study_Hours=c(2.0,3.5,1.5,4.0,2.8,3.0),
  Attendance=c(78,90,70,95,85,92),
```

```
  Math_Score=c(62,80,55,90,72,82),
  Science_Score=c(65,85,58,92,74,86),
  Exam_Date=as.Date(c("2025-01-10","2025-01-10","2025-02-12","2025-02-12","2025-03-15","2025-03-15"))
)
hist(students$Math_Score, col="lightblue",
     main="Histogram of Math Scores", xlab="Math Score")
boxplot(Science_Score ~ Gender, data=students,
        col=c("orange","pink"), main="Science Scores by Gender",
        xlab="Gender", ylab="Science Score")
```

Output:



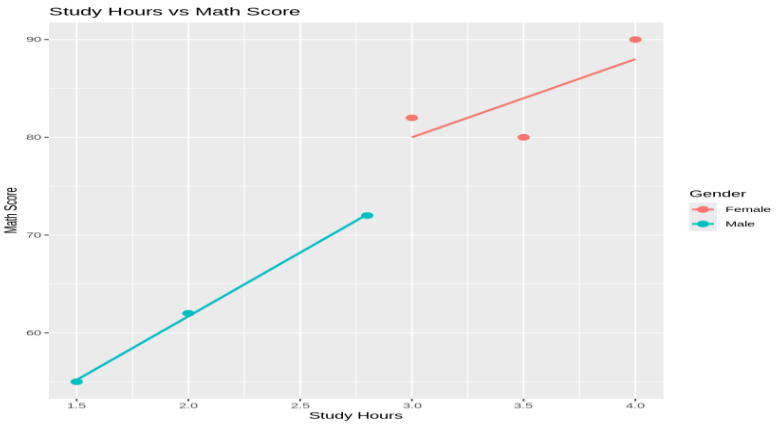
2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

Program:

```
# Dataset
students <- data.frame(
  StudentID=c("S01","S02","S03","S04","S05","S06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Study_Hours=c(2.0,3.5,1.5,4.0,2.8,3.0),
  Attendance=c(78,90,70,95,85,92),
  Math_Score=c(62,80,55,90,72,82),
  Science_Score=c(65,85,58,92,74,86),
```

```
  Exam_Date=as.Date(c("2025-01-10","2025-01-10","2025-02-12","2025-02-12","2025-03-15","2025-03-15"))
)
# Scatter plot with regression line
library(ggplot2)
ggplot(students, aes(x=Study_Hours, y=Math_Score, color=Gender)) +
  geom_point(size=3) +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Study Hours vs Math Score",
       x="Study Hours", y="Math Score")
```

Output:



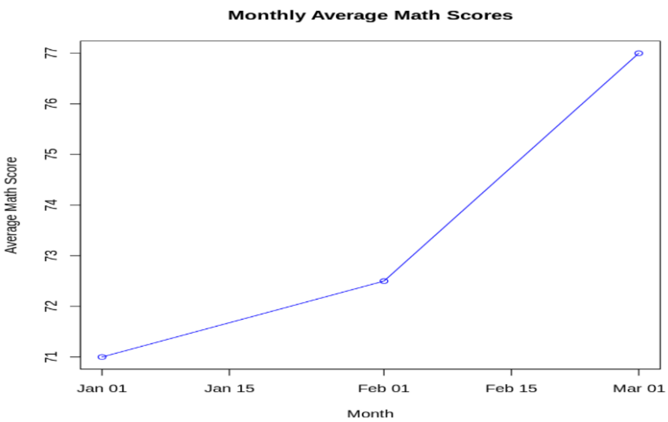
3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

Program:

```
# Dataset
students <- data.frame(
  StudentID=c("S01","S02","S03","S04","S05","S06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Study_Hours=c(2.0,3.5,1.5,4.0,2.8,3.0),
  Attendance=c(78,90,70,95,85,92),
  Math_Score=c(62,80,55,90,72,82),
  Science_Score=c(65,85,58,92,74,86),
```

```
  Exam_Date=as.Date(c("2025-01-10","2025-01-10","2025-02-12","2025-02-12","2025-03-15","2025-03-15"))
)
students$Month <- format(students$Exam_Date, "%Y-%m")
avg_scores <- aggregate(Math_Score ~ Month, students, mean)
# Plot with factor x-axis
plot(avg_scores$Month, avg_scores$Math_Score, type="o", col="blue",
     main="Monthly Average Math Scores", xlab="Month", ylab="Average Math Score")
# Moving average smoothing
library(zoo)
ma <- rollmean(avg_scores$Math_Score, 2, fill=NA)
lines(1:nrow(avg_scores), ma, col="red", lwd=2)   # use index for x
```

Output:



SET 27. Patient Health Risk Analysis

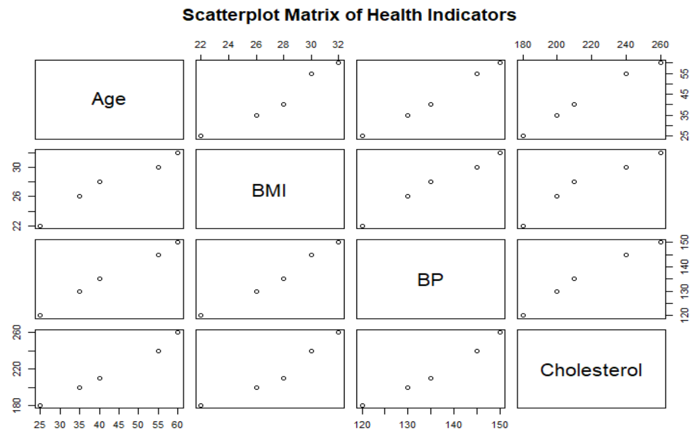
1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol.Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

Program:

```
patients <- data.frame(
  PatientID=c("P1","P2","P3","P4","P5"),
  Age=c(25,40,55,35,60),
  BMI=c(22,28,30,26,32),
  BP=c(120,135,145,130,150),
  Cholesterol=c(180,210,240,200,260)
```

```
)
pairs(patients[,c("Age","BMI","BP","Cholesterol")],
      main="Scatterplot Matrix of Health Indicators")
```

Output:

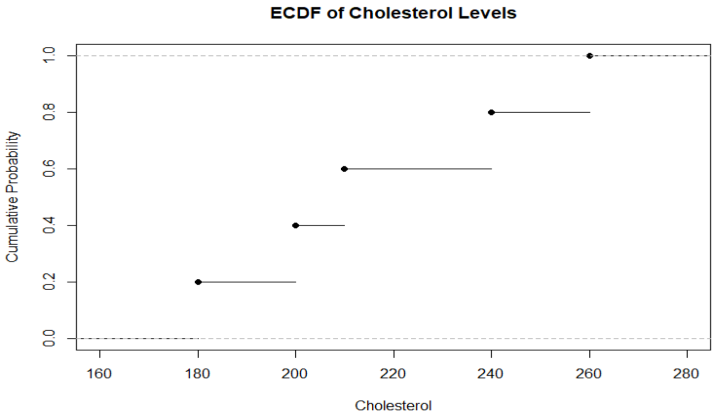


2.Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

Program:

```
patients <- data.frame(
  PatientID=c("P1","P2","P3","P4","P5"),
  Age=c(25,40,55,35,60),
  BMI=c(22,28,30,26,32),
  BP=c(120,135,145,130,150),
  Cholesterol=c(180,210,240,200,260)
)
# Q-Q Plot
qqnorm(patients$Cholesterol, main="Q-Q Plot of Cholesterol")
qqline(patients$Cholesterol, col="red")
# ECDF
plot(ecdf(patients$Cholesterol),
     main="ECDF of Cholesterol Levels",
     xlab="Cholesterol", ylab="Cumulative Probability")
```

Output:

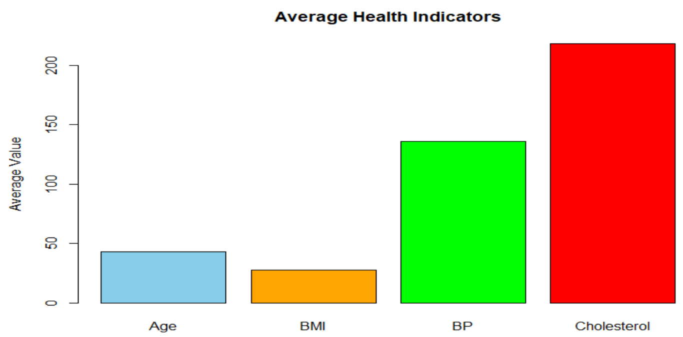


3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

Program:

```
patients <- data.frame(
  PatientID=c("P1","P2","P3","P4","P5"),
  Age=c(25,40,55,35,60),
  BMI=c(22,28,30,26,32),
  BP=c(120,135,145,130,150),
  Cholesterol=c(180,210,240,200,260)
)
avg_values <- colMeans(patients[,c("Age","BMI","BP","Cholesterol")])
barplot(avg_values, col=c("skyblue","orange","green","red"),
        main="Average Health Indicators",
        ylab="Average Value")
```

Output:



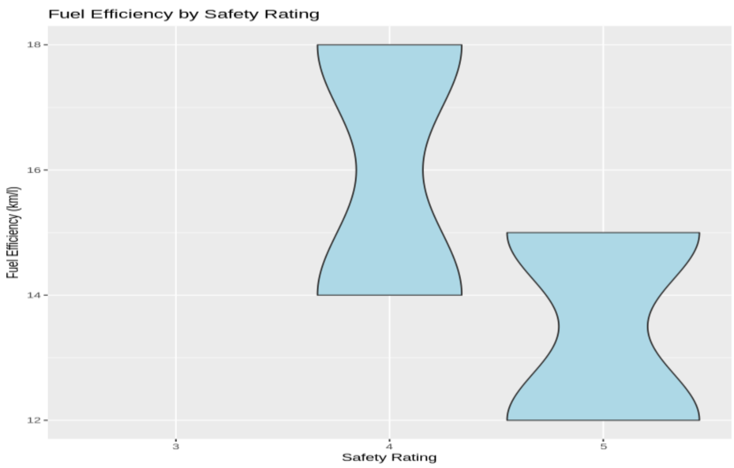
28. Vehicle Performance Analysis

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

Program:

```
vehicles <- data.frame(
  VehicleID=c("V1","V2","V3","V4","V5"),
  Engine_Size=c(1.5,2.0,3.0,2.5,1.8),
  Horsepower=c(110,150,250,200,130),
  Fuel_Efficiency=c(18,15,12,14,17),
  Top_Speed=c(180,200,250,220,190),
  Safety_Rating=c(4,5,5,4,3)
)
library(ggplot2)
ggplot(vehicles, aes(x=factor(Safety_Rating), y=Fuel_Efficiency)) +
  geom_violin(fill="lightblue") +
  labs(title="Fuel Efficiency by Safety Rating",
       x="Safety Rating", y="Fuel Efficiency (km/l)")
```

Output:

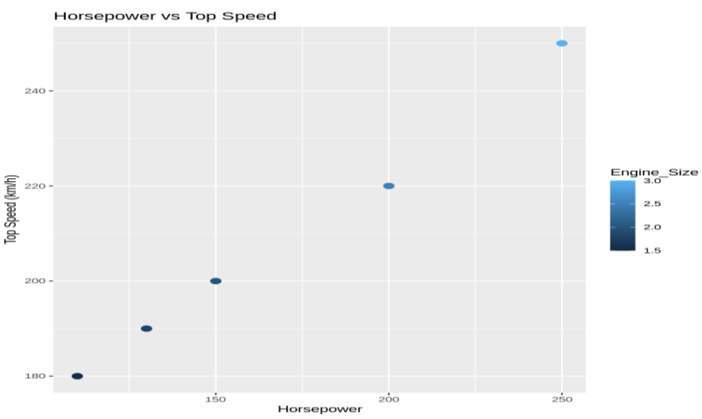


2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

Program:

```
vehicles <- data.frame(
  VehicleID=c("V1","V2","V3","V4","V5"),
  Engine_Size=c(1.5,2.0,3.0,2.5,1.8),
  Horsepower=c(110,150,250,200,130),
  Fuel_Efficiency=c(18,15,12,14,17),
  Top_Speed=c(180,200,250,220,190),
  Safety_Rating=c(4,5,5,4,3)
)
library(ggplot2)
ggplot(vehicles, aes(x=Horsepower, y=Top_Speed, color=Engine_Size)) +
  geom_point(size=3) +
  labs(title="Horsepower vs Top Speed",
       x="Horsepower", y="Top Speed (km/h)")
```

Output:



3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

Program:

```
vehicles <- data.frame(
  VehicleID=c("V1","V2","V3","V4","V5"),
  Engine_Size=c(1.5,2.0,3.0,2.5,1.8),
  Horsepower=c(110,150,250,200,130),
  Fuel_Efficiency=c(18,15,12,14,17),
  Top_Speed=c(180,200,250,220,190),
```

Safety_Rating=c(4,5,5,4,3)

)

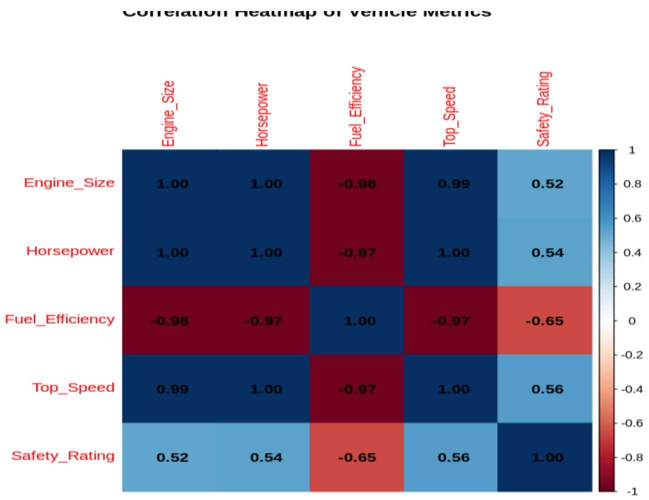
library(corrplot)

cor_matrix <-
cor(vehicles[,c("Engine_Size","Horsepower","Fuel_Efficiency","Top_Speed","Safety_Rating
")])

corrplot(cor_matrix, method="color", addCoef.col="black",

title="Correlation Heatmap of Vehicle Metrics")

Output:



SET 29 .Student Academic Performance

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

Program:

students <- data.frame(

Student_ID=c("S1","S2","S3","S4","S5"),

Age=c(19,21,20,22,23),

Study_Hours=c(12,8,15,10,7),

Attendance=c(90,70,95,85,60),

Test_Score=c(85,70,92,80,65),

Participation_Score=c(8,7,9,8,6)

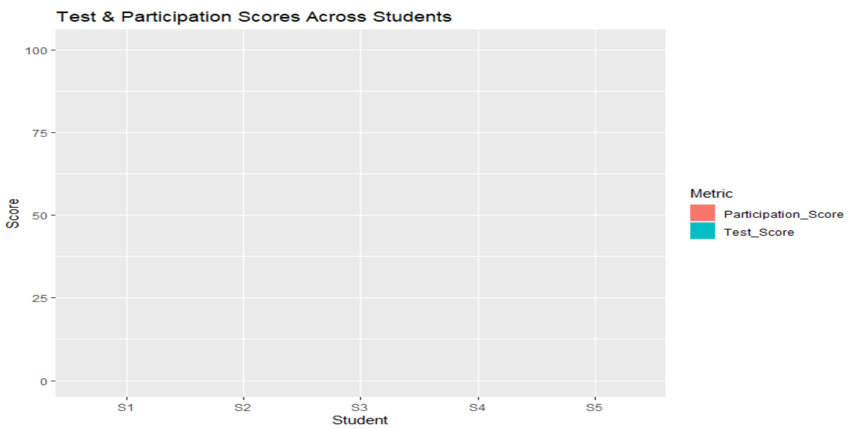
)

library(ggplot2)

library(tidyr)

```
students_long <- pivot_longer(students, cols=c(Test_Score, Participation_Score),
                              names_to="Metric", values_to="Score")
ggplot(students_long, aes(x=Student_ID, y=Score, fill=Metric)) +
  geom_area(stat="identity", position="stack") +
  labs(title="Test & Participation Scores Across Students",
       x="Student", y="Score")
```

Output:



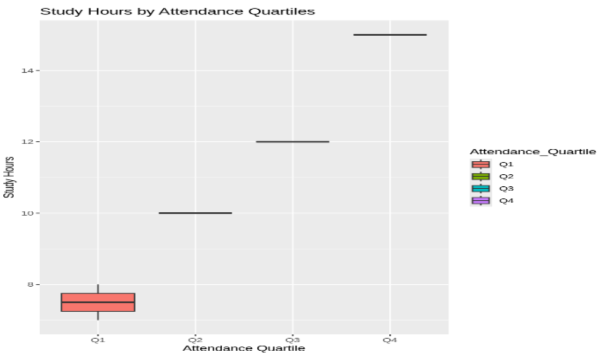
2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

Program:

```
students <- data.frame(
  Student_ID=c("S1","S2","S3","S4","S5"),
  Age=c(19,21,20,22,23),
  Study_Hours=c(12,8,15,10,7),
  Attendance=c(90,70,95,85,60),
  Test_Score=c(85,70,92,80,65),
  Participation_Score=c(8,7,9,8,6)
)
# Create quartiles
students$Attendance_Quartile <- cut(students$Attendance,
                                    breaks=quantile(students$Attendance, probs=seq(0,1,0.25)),
                                    include.lowest=TRUE, labels=c("Q1","Q2","Q3","Q4"))
```

```
ggplot(students, aes(x=Attendance_Quartile, y=Study_Hours, fill=Attendance_Quartile)) +
  geom_boxplot() +
  labs(title="Study Hours by Attendance Quartiles",
       x="Attendance Quartile", y="Study Hours")
```

Output:

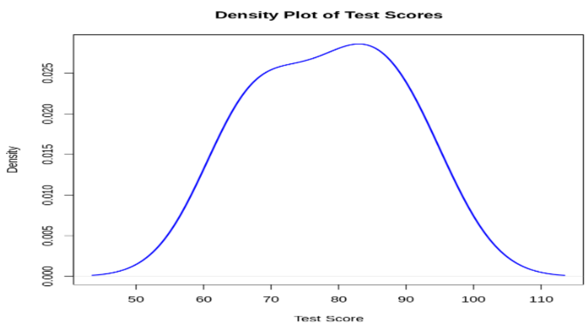


3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

Program:

```
students <- data.frame(
  Student_ID=c("S1","S2","S3","S4","S5"),
  Age=c(19,21,20,22,23),
  Study_Hours=c(12,8,15,10,7),
  Attendance=c(90,70,95,85,60),
  Test_Score=c(85,70,92,80,65),
  Participation_Score=c(8,7,9,8,6)
)
plot(density(students$Test_Score), col="blue", lwd=2,
     main="Density Plot of Test Scores", xlab="Test Score")
```

Output:



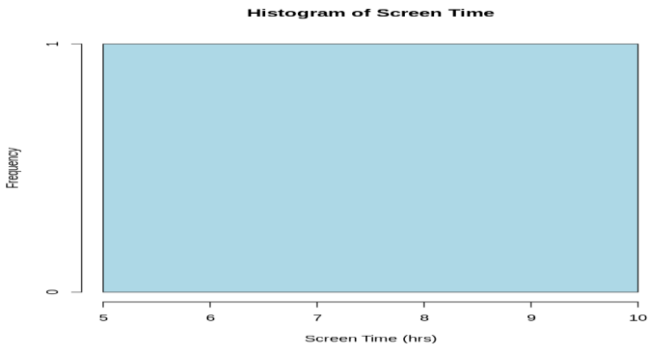
SET 30: Mobile_App_Usage

1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

Program:

```
usage <- data.frame(
  User_ID="U06", Gender="Female", Age=20,
  Screen_Time=5.4, App_Usage_Count=22, Data_Used=3.1,
  Satisfaction=4, Usage_Date=as.Date("2025-03-14")
)
hist(usage$Screen_Time, col="lightblue",
     main="Histogram of Screen Time", xlab="Screen Time (hrs)")
plot(density(usage$Screen_Time), col="red", lwd=2,
     main="Density Plot of Screen Time", xlab="Screen Time (hrs)")
```

Output:



2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

Program:

```
usage <- data.frame(
  User_ID="U06", Gender="Female", Age=20,
  Screen_Time=5.4, App_Usage_Count=22, Data_Used=3.1,
  Satisfaction=4, Usage_Date=as.Date("2025-03-14")
)
plot(usage$Screen_Time, usage$Data_Used,
     xlab="Screen Time (hrs)", ylab="Data Used (GB)",
     main="Data Used vs Screen Time", pch=19, col="darkblue")
```

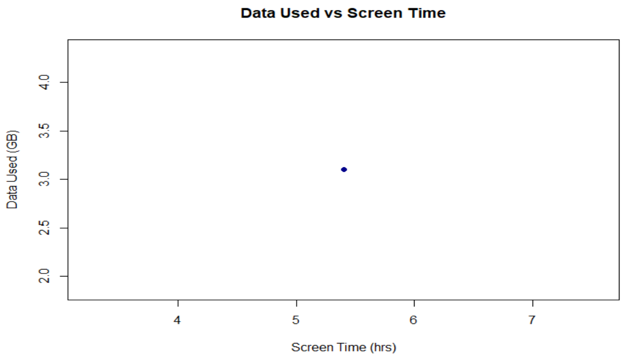
Correlation

cor(usage$Screen_Time, usage$Data_Used)

Trend line

abline(lm(Data_Used ~ Screen_Time, data=usage), col="red")

Output:



3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

Program:

```
usage <- data.frame(
  User_ID="U06", Gender="Female", Age=20,
  Screen_Time=5.4, App_Usage_Count=22, Data_Used=3.1,
  Satisfaction=4, Usage_Date=as.Date("2025-03-14")
)
avg_satisfaction <- aggregate(Satisfaction ~ Gender, usage, mean)
barplot(avg_satisfaction$Satisfaction, names.arg=avg_satisfaction$Gender,
        col="purple", main="Average Satisfaction by Gender",
        ylab="Satisfaction Score")
text(x=1, y=avg_satisfaction$Satisfaction, labels=avg_satisfaction$Satisfaction, pos=3)
```

Output: